



REVIEW

REVISED The Neuroscience of Marketing: Noninvasive Neuromarketing Methods for Understanding Consumer Choice and Priorities

[version 3; peer review: 2 approved with reservations, 2 not approved]

Previously titled: "Neuroscience of Marketing: Noninvasive Neuromarketing Methods for Understanding Consumer Choice and Priorities"

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Abstract

Neuromarketing and consumer neuroscience aim to enhance self-report and behavioral techniques by assessing neurophysiological and psychophysiological reactions that are frequently challenging to quantify. This study consolidates non-invasive tools frequently employed to examine consumer reactions to marketing stimuli, emphasizing the metrics each method assesses, the marketing inquiries it may reliably address, and the potential dangers of inference involved. We systematically compile evidence pertaining to constructs fundamental to consumer choice, including attention, affective arousal, memory encoding, and valuation, while comparing methodologies such as EEG, fMRI, TMS, Steady State Topography and MEG. The article proposes a method selection matrix that associates common research and applied inquiries (e.g., advertising attention dynamics, packaging assessment, pricing, and value indicators) with suitable measurement strategies, highlighting trade-offs concerning temporal/spatial resolution, ecological validity, cost, and interpretability. Finally, examine methodological challenges and ethical obligations. The scoping review concludes with research gaps

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where multimodal and more ecologically valid designs are most likely to advance the field.

Keywords

Neuroscience of Marketing, Neuromarketing, Consumer behaviour, Marketing, non-invasive techniques

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REVISED Amendments from Version 2

This revised version of the manuscript presents significant structural, conceptual, and analytical improvements compared to the previously published version. One of the primary changes is the shift from a descriptive framework to a more comparative and integrative approach. The new version introduces a method-selection matrix that allows for a more comprehensive comparison of non-invasive neuromarketing techniques, such as EEG, fMRI, MEG, TMS, eye-tracking, and Steady-State Topography, in relation to core consumer behavior constructs like attention, emotion, memory, and valuation. Additionally, we have provided a detailed analysis of the trade-offs among cost, ecological validity, and temporal versus spatial resolution, addressing gaps in the earlier manuscript.

The revised manuscript also features a more updated and expanded literature base, reflecting the latest trends and studies in the field. We have refined the language to enhance clarity and readability, addressing prior concerns regarding grammar, phrasing, and sentence structure. The conclusions section has been significantly revised to provide a clearer distinction between short-term applications and the long-term challenges facing neuromarketing, offering a more balanced and forward-looking perspective. Overall, the revisions ensure that the manuscript offers a more critical and coherent synthesis of the current state of neuromarketing research, with a focus on practical applications and future research directions, while adhering to higher standards of clarity, consistency, and scholarly rigor.

Any further responses from the reviewers can be found at the end of the article

Introduction

Marketing decisions today depend on consumers' quick, automatic responses to complex stimuli, such as short-form video ads, interactive shopping interfaces, and optimized product packaging (Lin et al., 2018; Morin, 2011). Traditional Self-reported methods such as surveys, interviews and focus groups are essential for understanding expressed beliefs and preferences; however, they are limited by recall bias, demand characteristics, and consumers' limited introspective access to the cognitive and emotional processes influencing attention, emotion, and decision-making (Pradeep et al., 2022). Consequently, essential elements of consumer decision-making frequently elude access by introspective methods alone.

Consumer neuroscience, and its applied extension neuromarketing, has emerged over the past two decades as a complementary framework aimed at addressing these limitations by integrating behavioral paradigms with noninvasive neurophysiological and psychophysiological measurements (Lin et al., 2018; Clithero, Carter & Huettel, 2024). Techniques such as electroencephalography (EEG), functional magnetic resonance imaging (fMRI), functional near-infrared spectroscopy (fNIRS), eye tracking, and peripheral autonomic measures enable the examination of attention allocation, memory encoding, affective arousal, and valuation as they unfold in real time (Rawnaque et al., 2020). These methods are increasingly adopted in marketing research as they reveal consumer responses that often occur outside conscious awareness, providing time-sensitive, objective indicators of the underlying processing (Mohd Isa & Anuar, 2024). Importantly, neuromarketing is not positioned as a replacement for traditional methods, but rather as a complementary approach that enhances explanatory power through triangulation integrating subjective reports, observed behavior, and physiological responses (Morin, 2011).

Despite its growth and increasing visibility, neuromarketing research still faces conceptual, methodological, and ethical challenges that hinder its overall impact. Previous reviews have often followed a technique-by-technique approach, providing descriptive overviews of individual tools without offering sufficient synthesis across methods or marketing-relevant constructs (Clithero et al., 2024). Persistent concerns regarding reverse inference, ecological validity, reproducibility, and ethical governance further complicate interpretation and application (Fisher et al., 2010; Gonçalves et al., 2024). Addressing these gaps, the present review focuses explicitly on noninvasive neuromarketing methods and contributes by (i) proposing a clear typology of commonly used neurophysiological and psychophysiological measures, (ii) offering an integrative comparison linking key marketing questions such as attention capture, emotional engagement, memory formation, and valuation to appropriate tools and their limitations, and (iii) synthesizing evidence to distinguish what is currently robust from what remains uncertain. Through adopting this construct-centered and method-comparative perspective, the review aims to advance both scientific rigor and responsible application in contemporary marketing research (Lin et al., 2018; Rawnaque et al., 2020). The review demonstrates the neuroscience of marketing as a systematic, noninvasive method for clarifying consumer choice and priority development, illustrating how neurophysiological evidence, when meticulously integrated with marketing theory and ethical standards, can enhance comprehension of automatic and context-dependent decision-making processes in contemporary consumer settings.

Understanding the consumer behaviour through neuroimaging paradigm

Currently, various sophisticated technologies are employed to map neural responses related to consumer behavior and other cognitive functions. Among these, two primary neuroimaging techniques functional magnetic resonance imaging (fMRI) and positron emission tomography (PET) are commonly used in neuroscience to visualize brain activity by

assessing changes in blood flow. These methods are grounded in the neurophysiological principle that mental activity increases the demand for oxygen and glucose in activated brain regions a demand that is met through increased cerebral blood flow (Bault & Rusconi, 2020; Kim, H. E. et al. 2022; Wivedi & Sharma, 2024). fMRI detects brain activity by measuring changes in blood oxygen levels. When neurons become active, blood flow increases, delivering more oxygenated blood to the region. This influx reduces deoxygenated hemoglobin concentration, enhancing the signal and allowing high spatial resolution mapping of neural activity. The Positron emission tomography, on the other hand, involves the injection of radioactive tracers that emit positrons. These tracers such as oxygen-15 and fluorodeoxyglucose (18F-FDG), accumulate in regions of the brain exhibiting increased neural activity, which corresponds to elevated blood flow (Rawnaque, Rahman, & Anwar, 2020; Alsharif & Mohd Isa, 2024). The resulting radiotracer emissions are detected by PET scanners to produce high-resolution images that reflect patterns of regional glucose metabolism an essential marker of brain function. Alterations in glucose metabolism often indicate underlying neurological or cognitive conditions. Rong, Lulong & Zhihua (2024) observed that medical imaging technologies are advancing rapidly, with molecular imaging emerging as a particularly promising field. The novel innovation in this area have sparked interest not only in clinical neuroscience but also in interdisciplinary applications such as neuromarketing. These innovations offer deeper insights into brain function and cognitive processes in consumer behavior (Guixeres et al., 2017; Garczarek-Bak, 2019; Russo, Clement, Jin, Liu, & Zito, 2022). Despite the availability of various neuroimaging methods, PET and fMRI remain the most prominent tools specifically utilized in neuromarketing research due to their ability to directly measure brain activity associated with emotional and decision-making processes.

Neuromarketing seeks to uncover the subconscious neural mechanisms influencing consumer behavior. Unlike traditional marketing approaches that rely on self-reported data, neuromarketing offers deeper insights into decision-making by analyzing brain activity and non-conscious responses. As Nemorin et al. emphasize, the distinctiveness of neuromarketing lies in its ability to circumvent the conscious cognitive filters of consumers, instead accessing brain activity directly to derive more authentic insights (Goncalves, Hu, Aliagas & Cerdá, 2024; Romanowski, 2019). Over the past five years, the majority of neuromarketing research has concentrated on two primary categories of stimuli: product-related cues (with or without price information) and promotional materials. In these studies, the term “product” encompasses both physical items such as tasting a beverage and conceptual depictions, like three-dimensional visual images. Price matters are frequently introduced as a key experimental variable, either independently or in conjunction with product or promotional content (Nemorin, 2016; Taqwa et al., 2015). The pricing information significantly affects consumer decision-making by activating specific cognitive and emotional responses. Empirical findings based on event-related potentials (ERPs) suggest that price promotions can enhance the perceived value of certain high-priced yet affordable luxury products, thereby increasing consumers’ intentions to purchase (Nemorin, 2016; Gorin et al., 2025).

The brain centric techniques, especially those utilizing advanced neuroimaging tools, are transforming how brands decode and influence consumer behaviour in today’s attention-driven economy. Beyond merely supplementing traditional methods, these approaches reveal real-time neural responses associated with emotional engagement, brand perception, and decision heuristics offering marketers evidence-based pathways to optimize campaign effectiveness (Plassmann et al., 2019). The various application of advanced techniques allows marketers to pinpoint the neural correlates of attention, memory encoding, reward processing, and emotional engagement (Petrovic, Petersson, & Ingvar, 2021; Venkatraman et al., 2012; Hasnaoui, Benabdallah, & Djebbari, 2023). These approaches are not only scientifically rigorous but also practically valuable, enabling firms to refine product designs, price strategies, and promotional content in ways that resonate deeply with target audiences (Chen & Hsu, 2023). For instance, fMRI research has been used to anticipate the success of advertising campaigns and music singles before their public release, indicating the strong real world applicability of neuromarketing (Berns & Moore, 2012; Wivedi, Sharma, 2024). While concerns about ethical application and methodological rigor are valid, the field has made considerable strides in standardizing protocols and addressing biases. As such, leveraging diverse and validated neuromarketing techniques remains a promising and legitimate pathway to achieving strategic marketing goals provided that methods are carefully selected and ethically implemented (Rawnaque, Rahman & Anwar, 2020; Morin, 2011). Studies reveal that consumer attitudes and behavioural responses are shaped by unconscious neural developments, often inaccessible through self-reports (Hasnaoui, Benabdallah, & Djebbari, 2023). The advance configured techniques like fMRI, EEG and other which enhance prediction accuracy for campaign effectiveness. These intellectual insights enable marketers to tailor strategies more precisely, bridging gaps left by traditional methods and improving engagement consequences.

Conventional to sophisticated insights of neuromarketing

The conventional approach to market research and data collection, aimed at fulfilling business requirements, proves inadequate in delivering the desired outcomes. Consequently, the adoption of novel techniques and methodologies for acquiring reliable evidence has become authoritative. To offset the substantial costs of advertising campaigns often undertaken by companies to promote new products or increase sales of existing one’s financial returns or measurable

market impact are typically required to justify the investment (Şik & Soba, 2021; Yang, Zhang, Qian, & Wang, 2022). It is widely acknowledged that the success of advertisements cannot be fully captured through verbal self-reports alone. Attempting to interpret emotional responses purely through cognitive or linguistic frameworks often leads to incomplete or misleading conclusions (Plassmann et al., 2015). Emotions are not solely grounded in language they are complex affective states that may elude conscious articulation. As such, accurately assessing emotional engagement requires moving beyond verbal expression to incorporate neurocognitive and physiological measures, which offer deeper insights into how consumers truly respond to advertising stimuli. Furthermore, such evaluations are subject to criticism for failing to capture the relationship between the influence of advertisements and subsequent consumer behavior (Romanowski, 2019). Consumer preferences, decisions, and choices are frequently influenced through Neuromarketing techniques. Despite businesses being hesitant to disclose the utilization of Neuromarketing tactics to enhance their marketing efforts, there have been a limited number of extensively documented research studies in this field. Various studies have indicated that the presence of attractive celebrities in advertisements stimulates specific brain regions, leading to the recognition of the product and the development of inner trust (Hubert & Kenning, 2008; Khushaba et al., 2019). In addition, customers are inclined to associate specific advertising elements with certain situations who respond strongly and make purchases accordingly. These studies enable businesses to shape, modify, and select advertisements in a manner that effectively resonates with and remains in the memory of their clients. The Specific marketing allows businesses to identify the neurological factors that consumers are engaged with in advertising. This provides a deeper understanding of how language, visuals, sound effects, and music can be utilized to create more compelling advertisements for their target audience (Gholami, 2024). Marketing managers have access to two sources of information that are crucial for making decisions: conventional research techniques and Neuromarketing techniques. The use of psychophysiological estimations allows participants to form preferences and make decisions through conscious thought, as opposed to relying on self-reported estimations of emotional states and unconscious cognitive processes that influence consumer responses in marketing studies. In other words, this approach goes beyond conscious aspects of human behavior and can identify influences on behavior even when consumers are not aware of them (Zaltman, 2003; Ko, Kim, & Lee, 2021). This made significant contribution to consumer research, considering that approximately 95% of our cognitive and emotional processing occurs at unconscious levels.

Currently, state-of-the-art eye-tracking systems and electrooculography (EOG) a technique that records eye movements by measuring the electrical potential between the front and back of the eye enable the detection of subtle changes in visual attention and ocular behavior. These technologies offer high temporal resolution, often within just a few milliseconds, specifically in tracking eye movement dynamics allowing researchers to precisely map attention patterns in real time (Duchowski, 2007; Khairunnisa & Sari, 2023). This is particularly useful for studying the effects of momentarily stimulating marketing stimuli and monitoring transient shopping behavior without impeding the decision-making process (Solnais, Andreu-Perez, Sanchez-Fernandez & Andreu-Abela, 2013). Electroencephalography (EEG) can be integrated with other advanced real-time determination techniques to provide additional insights into customer attention and interest (Hasnaoui, Benabdallah, & Djebbari, 2023). It can also be utilized in combination with other high time resolution technologies to provide more comprehensive information about viewer attention and arousal during advertising. For example, by synchronizing data from EEG recordings, it is possible to identify the precise focus of the member's attention at a given moment in time, as well as the specific elements of the scene that triggered activation of the left frontal hemisphere (Nemorin, 2016). The overall benefits of these technologies contribute to the increasing popularity of Neuromarketing and consumer neuroscience for advertising testing. Neuromarketing utilizes state-of-the-art resources in brain scanning to understand the customer's purchasing behavior. Schneider & Woolgar (2012) assert that Neuromarketing is the latest tool employed by marketing researchers to comprehend consumer behavior. In fact, understanding consumer behavior, particularly the decision-making processes involved in purchasing, emerges as one of the most frequently cited objectives in neuromarketing research (Eser et al., 2011; Suhendra, Hermita, & Darmayantie, 2015; Yang, Zhang, Qian, & Wang, 2022). Lee et al. (2007) also states that Neuromarketing has become a popular technology for determining the likelihood and non-probability of purchasing decisions, which has also been identified as a means of shaping companies' marketing strategies (Lee et al., 2007; Eser et al., 2011, Nemorin, 2016). Additionally, it helps to eliminate elements that should not be present in the communication, such as elements that lead to consumer aversion to the products. It also assists in the selection of visual and auditory features, as well as the timing and choice of appropriate media. Neuromarketing also has the ability to identify consumers' needs and, consequently, develop more meaningful and comprehensive consumer marketing strategies.

Non-invasive phenomena for understanding consumer behaviours

Advances in neuroimaging and cognitive neuroscience have facilitated the use of non-invasive techniques such as EEG, fMRI, MEG, and eye-tracking, among other biometric tools to decode consumer behavior with remarkable precision. These tools reveal unconscious emotional and cognitive processes, complement traditional methods by providing deeper insights, and establish a scientific foundation for evidence-based marketing interventions (Khushaba et al., 2019; Yang, Zhang, Qian, Wang, 2022; Hsu & Yoon, 2023). These tools reveal unconscious emotional and cognitive processes,

complements deeper insights than traditional methods and establishing a scientific foundation for evidence-based marketing interventions (Khushaba, et al., 2019; Yang, Zhang, Qian, & Wang; 2022; Hsu & Yoon, 2023). Central to this shift is the application of non-invasive brain imaging techniques that offer a powerful means for locating neural activity in response to marketing stimuli, allowing researchers and practitioners to decode how consumers prioritize, evaluate, and select products in complex environments. Among the most utilized non-invasive techniques, EEG stands out for its temporal resolution, portability, and cost-efficiency (Raiesdana & Mousakhani, 2022; Vecchiato et al., 2011), argue that EEG allows detailed observation of cognitive and emotional processing through measurements like event-related potentials (ERPs) and frontal asymmetry (Vecchiato et al., 2011; Mileti, Guido, & Prete, 2016). Increased activity in the left prefrontal cortex is linked to approach behavior and positive evaluations, making it a key biomarker for consumer preferences. As point out by Khushaba et al. (2013), EEG is particularly effective in contexts where consumers' conscious reasoning does not fully account for their final purchase decisions such as during exposure to emotionally charged advertisements or branding content.

Further expanding the scope of consumer behavior analysis, explore the theoretical and ethical boundaries of neuromarketing in comparison to "mind-reading." They emphasize that while non-invasive methods cannot access private thoughts, they can reliably capture automatic reactions that correlate with preference and motivation (Booth & Freeman, 2014). In particular, neurophysiological responses to product packaging, advertisement narrative arcs, or price visibility provide marketers with cues about what drives attention and valuation, thereby aiding in more precise campaign tailoring. Complementing the neurophysiological focus, Medina et al. (2020) employed neuroimaging tools to compare prosocial and non-prosocial consumer responses during pricing decisions. This study found that the insular and prefrontal regions show differentiated activation depending on ethical and prosocial dispositions. The implication here is that consumer priorities are not merely rational or economic but also neurologically modulated by identity-related values and social cognition, and these can be decoded using non-invasive methods like fMRI and EEG in combination.

An emerging and increasingly significant development within neuromarketing is the use of real-time neural data to drive highly personalized marketing strategies, which Mileti et al. (2016) describe as an advanced form of neuroadaptive marketing a specialized subset that dynamically tailors advertising content based on immediate neural responses. The neuroscience technique leverages consumer micro-signals, such as moment-to-moment emotional fluctuations detected by facial coding or eye-tracking, in conjunction with EEG signals to anticipate behavioral outcomes (Mileti et al., 2016; Yang, Zhang, Qian, & Wang, 2022). This hyper-individualized approach has profound implications for e-commerce and dynamic pricing models, allowing for the adaptation of product offers in milliseconds based on neural correlates of interest or aversion. The broader economic relevance of integrating neuromarketing into innovation management becomes particularly apparent when addressing uncertain consumer preferences in new product development (Romanowski, 2019). Further, non-invasive tools allow for early testing of prototypes, packaging, or pricing strategies, thereby reducing market failure risk. The strategic integration of neuromarketing within product innovation pipelines represents a methodological leap, embedding consumer intuition directly into corporate decision-making. The extensive non-invasive neuromarketing tools provide a scientifically grounded and ethically feasible method for uncovering the latent processes that shape consumer priorities. They circumvent the biases inherent in self-reported data, provide an objective basis for segmentation, targeting, and positioning. However, minor challenges remain in standardizing protocols, ensuring data interpretability, and maintaining consumer consent and transparency. Nonetheless, the field's trajectory supported by a growing body of evidence signals a development toward becoming an indispensable facet of modern consumer insight generation.

Neuroscience approaches for mapping consumer behaviour and choices

Traditional marketing tools often fail to capture the subconscious cognitive and emotional mechanisms that shape consumer behavior (Pradeep et al., 2022; Genevsky & Knutson, 2020). In contrast, neuromarketing leverages advanced non-invasive techniques such as EEG, fMRI, MEG etc. to observe neural responses in real-time, without requiring conscious effort from participants. These approaches allow researchers to pinpoint brain regions activated during brand exposure, decision-making, and emotional engagement, thus offering granular insights that surpass those obtained from self-reported data. Recent studies demonstrate that these neuroscientific methods significantly enhance the predictive validity of consumer responses, enabling marketers to design more effective strategies based on neurobiological evidence (Morin et al., 2023).

Neuromarketing is a highly influential marketing strategy in the contemporary world. The possibility and the impact of these techniques investigated using various methods and approaches (Stanton et al., 2017). The effective new methods are rooted in neuroscience, and neuroimaging techniques play a crucial role in testing hypotheses, enhancing existing knowledge, and examining the effects of marketing stimuli on consumers' brains. Neuro marketing strategies can be categorized into three broad categories: recording metabolic activities in the brain, recording electrical activity in the

brain, and which exploring brain activity without recording. Each of these categories differs in nature and serves as significant parameters for exploring the intentions, thoughts, hesitations, and other influential factors within the marketing industry.

Electroencephalography

Electroencephalography (EEG) is a highly effective and cost-efficient technique for investigating the dynamics between brain activity and behavior (Bell & Cuevas, 2012). Notably, the examination of various spectral bands, including Delta (0– Hz), Theta (3–7 Hz), Alpha (8–12 Hz), Beta (13–30 Hz), and Gamma (30–0 Hz), has been employed to analyze consumers' cognitive and affective processes in response to marketing stimuli (Mostafa, 2012). The versatility of this method allows researchers to explore an extensive range of fields, enabling a comprehensive understanding of the temporal sequencing of neural events (Light et al., 2010). The recording of EEG signals is conducted by placing multiple electrodes on the participant's scalp in order to enhance the conduction of impulses to the electrodes. The electric current generated by the typical human brain is on the order of a few microvolts. These voltage fluctuations are a result of ionic currents flowing between the brain and nerve cells. Nerve cells communicate with each other through electrical impulses (Mileti, Guido, & Prete, 2016; Yang, Zhang, Qian, & Wang, 2022; Hsu & Yoon, 2023). The electrical potential detected by an individual neuron is extremely weak and therefore not detectable. Each electrode reflects the combined activity of thousands or millions of neurons with the same spatial orientation, thus the EEG measures the overall synchronous activity of a large number of neurons present in the brain. It is highly valuable in evaluating emotional stability and monitoring emotional conditions (Hsu & Yoon, 2023; Kline, 2000). This device is a portable tool that aids in assimilating and synchronizing stimuli. The only limitation of this method is that it can only record apparent electrical signals and cannot differentiate deep brain structures or exchange information. Consequently, the EEG provides greater temporal resolution and less spatial resolution in the field of neuroscience. The electroencephalogram technology, frequently utilized in neuromarketing, offers insightful data on brain activity and is both portable and reasonably priced.

Advanced wireless electroencephalogram device represents a significant breakthrough in neuromarketing to enabling real-time, mobile, and ecologically valid monitoring of brain activity in naturalistic environments. These devices offer high temporal resolution and improved comfort, allowing researchers to track consumers' neural responses to advertisements, retail layouts, and digital content outside laboratory constraints (Ma et al., 2022; Luis-Alberto, & Sanchez-Fernandez, 2022). The effectiveness in capturing emotional engagement, cognitive load, and attention metrics during real-world consumer experiences. As wearable EEG technology becomes more sophisticated, it opens new possibilities for continuous consumer insight, adaptive advertising, and personalized marketing strategies based on neurophysiological feedback.

Functional magnetic resonance imaging

Functional Magnetic Resonance Imaging (fMRI) is widely regarded as one of the primary methods for investigating brain activity. This technique enables the analysis of intricate and minute brain structures; however, it is more expensive than many other methods. Moreover, the 6–10 second delay in recording neural activity represents a limitation when measuring responses to marketing stimuli (Ariely & Berns, 2010). There are two main techniques used in functional MRI: the Blood Oxygen Level Dependent (BOLD) technique and the dynamic exogenous technique (Hare et al., 1998). The BOLD technique is preferred as it does not require intravenous contrast. Chow et al. (2017) highlight that the majority of fMRI studies utilize BOLD contrast imaging, which involves mapping active regions of the brain based on changes in blood oxygen. Blood flow in the brain is locally regulated in response to oxygen and carbon dioxide levels in cortical tissue. This method has garnered significant attention in the field of Neuromarketing. While fMRI provides excellent spatial resolution, its temporal resolution is relatively poor. In Neuromarketing, fMRI is employed to measure various factors such as memory encoding, emotional valence, sensory perception, brand loyalty and trust, brand preference, and brand recall.

Recent advancements have leveraged fMRI to study neural valuation systems particularly activity in the ventromedial prefrontal cortex (vmPFC) and nucleus accumbens as reliable predictors of purchasing intention and product desirability (Enax et al., 2015; Plassmann et al., 2021). Furthermore, emerging fMRI-based research explores the brain's response to sustainability messaging and ethical branding, revealing heightened activation in areas linked to moral reasoning and social cognition, such as the temporoparietal junction and dorsomedial prefrontal cortex (Karmarkar, Bollinger, 2022). This study provides a new dimension to understanding consumer behaviour, where decision-making is influenced not only by price and utility, but also by values, emotional resonance, and ethical alignment. As fMRI continues to evolve with improved models and machine learning integration, it offers deeper insights into the subconscious processes driving consumer preferences in increasingly complex marketplaces.

Transcranial Magnetic Stimulation

Transcranial Magnetic Stimulation (TMS) is a non-invasive neuroscience technique used to stimulate targeted areas of the cerebral cortex which enables researchers to establish causal links between specific brain regions and behavioural responses to inducing temporary changes in neural activity. The device placed to the scalp, TMS generates a magnetic field that stimulates underlying brain tissue, unsettling or enhancing cortical processing in localized areas (Sliwinska, Vitilo & Devlin, 2014). This permits for precise manipulation of cognitive functions and provide insights into the timing and role of neural processes in real-time decision-making. In addition, consumer research, TMS provides a powerful technique to explore how the brain responds to marketing stimuli. By selectively activating or inhibiting regions associated with emotion, attention, and decision-making, researchers can examine how these changes affect purchasing attitudes and behavior. Recent studies using TMS on the dorsolateral prefrontal cortex (DLPFC) have shown that modulating this region can significantly alter consumer assessment and risk preferences during purchasing decisions (Gupta et al., 2021). Moreover, this technique has been effectively used to study the influence of affective priming on product judgments, indicating that temporary disruption of emotion-processing circuits can modulate brand attitudes and consumer trust (Zhang & Li, 2022). Research findings support the growing role of TMS in neuromarketing, where it helps decode, the neural mechanisms underlying consumer attention, preference formation, and decision-making, enabling marketers to develop empirically informed engagement strategies.

Indeed, TMS can directly influence consumption-related behaviours, such as delaying gratification or shifting preference toward serviceable versus hedonic products (Ramsoy et al., 2021). Furthermore, emerging TMS applications in neuromarketing explore the modulation of empathy and social cognition through ethical consumption decisions, revealing that stimulating the right temporoparietal junction can alter perceived brand authenticity and trust (Cian et al., 2022). Through effectively use these techniques is not merely a validation tool for neural correlates but a mechanism for probing and reshaping the cognitive architecture of consumer thought itself. This opens promising avenues for designing interventions in areas such as compulsive buying, financial risk behavior, and ethical advertising.

Steady State Topography

Steady-State Topography (SST), introduced by Richard Silberstein in 1990, is a cutting-edge neurophysiological method for mapping neural activity in response to external stimuli. Initially developed for cognitive neuroscience, this device supports significant applications in neuromarketing and consumer neuroscience, especially in evaluating consumer perception, emotive engagement, and brand communication (Maran, Moonisha, Patel, Mari Muthu, & Anbazhagan, 2021). In SST experiments, participants are exposed to audiovisual stimuli while a dim sinusoidal flicker is presented in their peripheral vision, causing a steady-state visually evoked potential (SSVEP). This neural response is measured across cortical areas to assess variations in cognitive and emotional processing. The specific concern of SST particularly, valuable in neuromarketing is its sensitivity to changes in latency the delay between stimulus and neural response which reflects long-term memory encoding, sustained attention, and emotional intensity (Silberstein & Nield, 2021). Through analysing latency shifts and cortical activation patterns over time, SST enables a more nuanced understanding of how consumers perceive advertising messages, process branding cues, and form durable attitudes toward products. Steady-State Topography based new research demonstrated its ability to predict advertising effectiveness and product recall with high precision, making it an advanced tool for capturing subconscious consumer behavior in real-world media environments.

Magnetoencephalography

Magnetoencephalography (MEG) is a technique that measures brain activity by detecting potential magnetic fields on the scalp using sensitive detectors housed in a helmet placed on the head. Unlike other methods, MEG is not influenced by tissue type, such as blood, brain matter, or bone, and can provide spatial and temporal resolution to determine the depth of a point in the brain. The MEGs are capable of monitoring population neuronal activity in the brain, but the costs associated with their use are significant due to the requirement for a magnetic field-free environment. This device records electrical activity in the brain, offering high temporal resolution to trace the consumers' responses to marketing stimuli in real time.

The Magnetoencephalography offer unique advantage in neuromarketing method through providing millisecond-level temporal resolution and superior spatial localization of brain activity compared to Electroencephalography, while remaining non-invasive unlike functional magnetic Resonance. This method is very magnificent to captures magnetic fields generated by neuronal activity which allowing precise tracking of consumer responses to marketing stimuli such as advertisements and product designs (Lee et al., 2021; Morinaga et al., 2022; Dikker & Pineda, 2023). Undouble this unique method effectively maps cognitive processes like brand recall, emotional arousal, and decision-making pathways. This real-time, high-fidelity insight supports profound consumer profiling and enhances predictive modelling in marketing strategies.

Conclusion

Neuromarketing is most valuable when used as a complement to established behavioral and self-report methods, rather than as a tool for directly ‘reading’ consumers’ minds. By integrating neurophysiological insights with traditional marketing research, neuromarketing provides a deeper understanding of unconscious consumer responses, such as emotional engagement, attention, and memory encoding, which are often inaccessible through conventional methods alone. Techniques like EEG, fMRI, and TMS offer real-time, objective data that can refine marketing strategies, enhancing the precision of advertisements, pricing, and product designs. However, while these techniques have proven effective in capturing the intricacies of consumer behavior, they also face notable challenges. These include limitations related to spatial and temporal resolution, as well as ethical concerns surrounding privacy, consent, and potential manipulation. The future of neuromarketing depends on addressing these issues through the development of multimodal, longitudinal, and cross-cultural research, which can provide a more holistic view of consumer behavior. Furthermore, aligning neuromarketing with ethical standards and marketing theory will be crucial in ensuring its responsible and sustainable application. Ultimately, the long-term success of neuromarketing hinges on balancing scientific rigor with ethical responsibility, offering marketers not just deeper insights into consumer behavior, but also a path to more effective and ethically sound marketing practices.

Data availability

All data are publicly available and cited in the manuscript. No new data were generated, and no DOI applies.

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Open Peer Review

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Singgih Santoso 

Universitas Kristen Duta Wacana Yogyakarta, Yogyakarta, Indonesia

Article makes use of non-invasive neuromarketing techniques to assess how consumers make purchasing decisions. Such technologies include EEG, fMRI, MEG, TMS, Steady-State Topography, eye-tracking and peripheral psychophysiology, and are related to the core constructs of attention, emotion, memory and valuation. And the article also includes a method-selection matrix which links common marketing-related questions (e.g. ad testing, packaging, pricing) with applicable techniques and describes the trade-offs involved for both cost and temporal/spatial resolution versus ecological validity.

comments

First, the scope and method of the review are still not clearly described. I would recommend that the authors state explicitly that this is a narrative/scoping review and briefly describe how they identified the relevant literature (e.g., the main databases searched, the years covered, the keywords used, and the basic criteria for inclusion). This will provide clarity regarding what is meant by “comprehensive” coverage and will respond to earlier concerns about the transparency of the review process.

Second, the method- selection matrix is a nice addition; however, the synthesis of the matrix can be improved. It would be helpful for readers if the authors provided some specific examples to help them understand why certain methods or combinations of methods are better suited for use than others (e.g., pre-testing TV ads, evaluating sustainable branding, optimizing pricing). In addition, it would be helpful for the authors to include a short subsection that summarizes what we know and don't know for each of the four main constructs (i.e., attention, emotion, memory, and valuation) at the end of the conclusions to help ground the conclusions in evidence from the literature review.

third, the lack of consistency in terms and concept relating to ‘non-invasive’ measures is still a major issue. Those who previously reviewed research in this area pointed out that PET is one example that has complicating factors. The author should clarify the use of the term “non-invasive” by developing a typology (metabolic imaging, electrophysiology, neuromodulation,

peripheral psycho-physiological, and behavioural / implicit).

Fourth, a welcome addition to the paper is the inclusion of discussion about methodical and ethical challenges; although they were not explored in detail. The paper could benefit from a more rigorous examination of the issues raised by breaking down the issues using examples of situations where neuromarketing outcomes could not predict reality or where low sample sizes/inverse inference have led to conclusions that have been overgrown. I also propose developing a checklist of ethical practices and principles for practitioners regarding consent, privacy, use of data, and limitations of neuroadaptive marketing and including them in a table in this article to increase the practical relevance of the article.

Overall recommendation

After reviewing the article and its revisions, I recommend it as approved, but with reservations. After clarifying their review method, providing more integration around their method selection matrix, strengthening their critique of methodology and ethical considerations, and improving the clarity of the language used to describe these issues, this paper will provide an important addition to the literature.

NOTE:

The Turnitin test shows that the article has a similarity score of 97% and an AI writing score of 57%, which proves that the article was generated by AI and contains plagiarism. In order for it to be published, please reduce the similarity score and ensure the AI writing score is BELOW 30%.

Is the topic of the review discussed comprehensively in the context of the current literature?

Partly

Are all factual statements correct and adequately supported by citations?

Yes

Is the review written in accessible language?

Partly

Are the conclusions drawn appropriate in the context of the current research literature?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Marketing

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Version 2

Reviewer Report 30 March 2026

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Ramachandran K K 

GRD Institute of Management, Coimbatore, India

The manuscript presents a narrative review of non-invasive neuromarketing methods employed to examine consumer behavior. It discusses key techniques such as EEG, fMRI, MEG, TMS, eye-tracking, and Steady-State Topography, and connects them to core constructs including attention, emotion, memory, and valuation. The authors seek to move beyond purely descriptive accounts by providing a more comparative and integrative framework that incorporates a method-selection matrix and an analysis of trade-offs among cost, ecological validity, and temporal versus spatial resolution. In addition, it outlines potential directions for future research, particularly emphasizing the importance of multimodal approaches and more ecologically valid study designs.

Strength:

1. The paper successfully links neuroscientific methods with core marketing constructs such as attention, emotion, memory, and valuation.
2. It incorporates recent literature, reflecting current developments in the field.
3. A well-written article that is appropriate to the journal.

Weakness:

1. The readability is partially effective as several sections contain errors in grammar and phrasing and are frequently dense and repetitive. This makes it difficult to follow the arguments.
2. Ensure consistency

Comments:

The manuscript addresses a relevant and emerging topic and offers a valuable integrative perspective on non-invasive neuromarketing methods. The effort to connect techniques with core consumer behavior constructs and to propose a comparative framework is a notable strength. To avoid redundancy, simplify the sentence structure to make it easier to follow.

The manuscript presents a meaningful contribution to the field and is suitable for indexing with a moderate revision addressing clarity, and consistency.

Is the topic of the review discussed comprehensively in the context of the current literature?

Yes

Are all factual statements correct and adequately supported by citations?

Yes

Is the review written in accessible language?

Partly

Are the conclusions drawn appropriate in the context of the current research literature?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Marketing, Management, AI, Retail Management

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Version 1

Reviewer Report 29 December 2025

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1. Is the topic of the review discussed comprehensively in the context of the current literature? (Partly)

The paper covers many key tools and cites a broad range of literature, including recent work on EEG, fMRI, MEG, TMS, eye-tracking, and neuromarketing ethics.

However, the discussion is mostly descriptive and technique-by-technique, with limited synthesis across studies, and there is no explicit method for selecting or delimiting the reviewed literature. Important dimensions such as cross-modal comparisons, application domains (e.g. branding, pricing, digital platforms), and known controversies (e.g. reverse inference, effect sizes, replicability) are touched on only briefly, if at all.

- Clarify the **scope and method** by state explicitly that this is a narrative/scoping review rather than a systematic review, and briefly describe how the authors identified and selected the literature
- Add at least one **integrative section or figure** (e.g. a table or matrix) that:
 - compares techniques on temporal/spatial resolution, cost, ecological validity, and typical marketing questions they address; and
 - summarises “what we know” after two decades of consumer neuroscience (e.g.

robust findings on value coding, attention, memory, and predictive validity for real-world outcomes).

- Expand the **“big picture”** integration in the conclusion: what are the main areas of consensus/disagreement in the literature, and where are the most promising gaps (e.g. multi-modal approaches, longitudinal designs, cross-cultural neuromarketing)?

2. Are all factual statements correct and adequately supported by citations?

Non-invasive” and invasive inconsistency. The title and abstract frame the review as focusing on *non-invasive* methods, listing PET alongside fMRI, EEG, etc. Later, the PET section correctly notes that PET involves injection of radioactive tracers and is therefore invasive and less commonly used. This is **internally inconsistent** and conceptually misleading. Either (a) remove PET from the “non-invasive” framing in the title/abstract and clearly state that PET is *minimally invasive* and now rarely used in consumer work, or (b) broaden the title/scope beyond “non-invasive” and clearly distinguish invasive vs non-invasive methods throughout.

The paper sometimes groups very different tools under “non-invasive neuromarketing techniques” (e.g. PET, TMS, implicit association tests, biometric sensors) without clearly distinguishing neuroimaging / electrophysiology from behavioural / psychophysiological measures.

Use a consistent typology (e.g. *metabolic imaging* (fMRI, PET), *electrophysiological* (EEG, MEG, SST), *neuromodulation* (TMS), *peripheral psychophysiology* (GSR, HR, facial EMG), *behavioural implicit measures*).

4. Are the conclusions drawn appropriate in the context of the current research literature?

However, the conclusions are quite general and do not always flow directly from the preceding sections. They would be more impactful if they more clearly summarised what the reviewed evidence actually shows and what remains uncertain.

Anchor conclusions to specific evidence. For example, summarise key robust findings (e.g. value coding in vmPFC/NAcc, predictive power of certain neural markers for campaign success, reliability of frontal alpha asymmetry for approach-avoidance tendencies).

Differentiate short-term promise vs long-term challenges

Is the topic of the review discussed comprehensively in the context of the current literature?

Partly

Are all factual statements correct and adequately supported by citations?

Partly

Is the review written in accessible language?

Yes

Are the conclusions drawn appropriate in the context of the current research literature?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Consumer Behavior, marketing, neuromarketing

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

Reviewer Report 11 December 2025

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Jesús Jaime Moreno Escobar 

ESIME - Escuela Superior de Ingeniería Mecánica y Eléctrica Unidad Zacatenco, Mexico City, Mexico

The proposal advocates for the use of non-invasive neuromarketing techniques—such as fMRI, EEG, PET, MEG, and eye-tracking—to understand the subconscious drivers of consumer behavior. This approach aims to overcome the limitations of traditional methods like surveys and focus groups by directly measuring brain activity and physiological responses.

Advantages:

1. **Access to Subconscious Processes:** The primary advantage is the ability to bypass the limitations of self-reporting. These techniques can reveal unconscious emotional, cognitive, and decision-making processes that consumers cannot or will not articulate, providing more authentic insights.
2. **Objective and Rich Data:** Neuroimaging provides objective, biological data on consumer responses to marketing stimuli (ads, products, brands). This allows for a deeper understanding of the neural mechanisms behind attention, emotional engagement, memory encoding, and preference formation.
3. **Improved Predictive Power:** Studies show that neural data can predict market outcomes, such as advertising success and product desirability, more accurately than traditional methods. This allows companies to refine product designs, pricing, and promotional content with greater confidence, potentially reducing market failure.
4. **Real-Time and High-Resolution Insights:** Techniques like EEG and MEG offer very high temporal resolution (milliseconds), allowing researchers to track consumer responses in real time. This is crucial for understanding the immediate impact of fast-paced marketing stimuli.
5. **Application to Innovation and Personalization:** Neuromarketing supports new product development by testing prototypes and can enable hyper-personalized, "neuroadaptive" marketing where content is dynamically tailored based on real-time neural feedback.

Disadvantages:

1. **High Cost and Complexity:** Many of these tools, particularly fMRI and MEG, are extremely expensive to acquire and operate. They also require highly specialized expertise in both neuroscience and data analysis, limiting their accessibility.
2. **Methodological Limitations:** Each technique has inherent trade-offs. fMRI has excellent

spatial resolution but poor temporal resolution and a high cost. EEG is portable and has great temporal resolution but poor spatial resolution and cannot probe deep brain structures.

3. **Ethical and Privacy Concerns:** The field raises significant ethical questions about "mind-reading," consumer manipulation, and data privacy. There is public apprehension about the potential for brands to exploit subconscious biases and vulnerabilities.
4. **Interpretation Challenges:** Linking specific neural signals to complex psychological constructs like "brand loyalty" or "purchase intent" is inferentially complex. The data requires careful interpretation, and standardized protocols are still evolving, which can lead to variability in results.
5. **Limited Adoption and Transparency:** The high barriers to entry have slowed widespread adoption in market research. Furthermore, businesses are often hesitant to disclose their use of neuromarketing, which fuels skepticism about its application.

Dear Editor,

Thank you for the opportunity to review this manuscript. I have evaluated it based on standard scholarly criteria. My overall recommendation is to **reject the manuscript in its current form, but with encouragement for a significant resubmission**. The manuscript addresses a topic of considerable interest but requires substantial revisions to meet the journal's standards. Below is my detailed evaluation.

1. Originality / Novelty

The topic of applying non-invasive neuromarketing techniques is inherently relevant. However, the manuscript's originality is limited. It functions primarily as a descriptive catalog of established methods (fMRI, EEG, PET, MEG) and their general advantages and disadvantages, rather than presenting a novel synthesis, a new conceptual framework, or a critical gap analysis that would distinguish it from other existing reviews in the field. The authors have not clearly demonstrated how their approach or synthesis offers a new perspective compared to the extant literature.

2. Significance of Content

The content is of significant potential interest to the fields of marketing, consumer behavior, and applied neuroscience. Understanding the subconscious drivers of consumer decision-making is a central challenge, and a well-executed review that critically appraises the value and limitations of these tools would be highly valuable. The potential for impact is present, but it is not fully realized in the current draft.

3. Quality of Presentation

This is a major weakness of the manuscript. The authors **do not use a consistently readable level of English**. Grammatical errors, awkward phrasing, and unclear syntax are prevalent throughout the text, which significantly hinders comprehension and detracts from the scholarly tone. Furthermore, the logical structure is problematic; the justification for the review and the clear articulation of the main problem it aims to solve are not effectively exposed, making the manuscript's core contribution difficult to discern.

4. Scientific Soundness

The scientific soundness is compromised by several factors. While the reference list is relevant, it is not sufficiently current, with only 45% of the 88 references from the last five years. This undermines the manuscript's claim to represent the state-of-the-art in a rapidly evolving field. More critically, the review lacks a balanced critical perspective. It lists disadvantages (e.g., cost, ethical concerns) but does not engage with them in a substantive way that would provide a rigorous appraisal of the field's validity and reliability. The

manuscript describes rather than critiques.

5. Interest to the Readers

A robust and well-written review on this topic would be of high interest to the diverse readership of *F1000Research*, including neuroscientists, marketing professionals, and behavioral economists. The subject matter aligns well with interdisciplinary research. However, the current presentation issues significantly reduce its accessibility and, therefore, its potential impact.

6. Overall Merit and Critical Concern

The manuscript has a strong foundational premise and covers a valuable topic. However, its current overall merit is low due to the critical issues in presentation, originality, and scientific rigor. The **originality report shows a 19% similarity index, which is acceptable, but also flags that 71% of the content has a high probability of being AI-generated.**

This is a severe concern. The stilted language, lack of a coherent narrative voice, and descriptive rather than critical analysis are consistent with this finding. For a scholarly publication, the intellectual synthesis and critical perspective must be demonstrably the product of the authors' expertise.

Recommendation:

I recommend **rejecting the current manuscript but encouraging a resubmission** after the authors undertake a thorough, top-to-bottom revision. The authors should:

6. Completely rewrite the manuscript to ensure clarity, fluency, and a strong, critical authorial voice, decisively addressing the concerns of AI-assisted writing.
7. Restructure the paper to clearly state its unique contribution, define a specific gap in the literature, and provide a compelling justification for the review.
8. Update the literature review significantly to include more recent publications (post-2019).
9. Move beyond description to provide a critical, balanced analysis that deeply engages with both the advantages and the methodological and ethical challenges of neuromarketing.
10. The core topic has great potential, and I would be interested in reviewing a fundamentally revised version that demonstrates genuine scholarly engagement and critical thought.

Is the topic of the review discussed comprehensively in the context of the current literature?

No

Are all factual statements correct and adequately supported by citations?

Yes

Is the review written in accessible language?

Yes

Are the conclusions drawn appropriate in the context of the current research literature?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Computer Vision, Data Analytics

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

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